

# FINANCIAL SALES ANALYSIS

## Internship Project Report

**Prepared By**

Skandashree R

### **Technologies Used**

Python (Pandas, NumPy)

MySQL Workbench

Microsoft Power BI

Microsoft Excel

## Acknowledgement

I sincerely thank my mentors, internship coordinators for their guidance and support throughout this project. This internship enhanced my practical knowledge of Python, SQL and Microsoft Power BI while strengthening my data analytics skills.

## Table of Contents

| Sl No | Topics               | Pages |
|-------|----------------------|-------|
| 1     | Executive Summary    | 1     |
| 2     | Introduction         | 1     |
| 3     | Problem Statement    | 1     |
| 4     | Objectives           | 1     |
| 5     | Dataset Description  | 1     |
| 6     | Methodology          | 2     |
| 7     | Python Data Cleaning | 3-7   |
| 8     | SQL Analysis         | 8-15  |
| 9     | Power BI Dashboard   | 16-20 |
| 10    | Business Insights    | 21-22 |
| 11    | Recommendations      | 23    |
| 12    | Future Scope         | 24    |
| 13    | Conclusion           | 24    |
| 14    | References           | 25    |

## **Executive Summary**

This report presents an end-to-end financial sales analytics solution. The workflow included data cleaning in Python, business analysis using SQL, and visualization using Microsoft Power BI. The objective was to transform raw sales data into meaningful business insights through a structured analytics process.

## **Introduction**

Data analytics helps organizations understand trends, monitor business performance, and support decision-making. This project demonstrates a complete analytics pipeline suitable for business intelligence applications.

## **Problem Statement**

Raw business data often contains missing values and requires preprocessing before analysis. The challenge was to clean the dataset, analyze sales performance, and create an interactive dashboard for stakeholders.

## **Objectives**

- Clean and validate the dataset.
- Handle missing values and verify duplicates.
- Perform SQL-based business analysis.
- Build an interactive Power BI dashboard.
- Present actionable business insights.

## **Dataset Description**

The financial sales dataset contains 700 records and 16 attributes such as Segment, Country, Product, Discount Band, Units Sold, Gross Sales, Sales, Profit, Date and Year.

## Financial Sales Analysis | Internship Project

| Segment    | Country     | Product   | Discount Band | Units Sold | Manufacturing Price | Sale Price | Gross Sales | Discounts | Sales    | COGS   | Profit   | Date      | Month Number | Month Name | Year |
|------------|-------------|-----------|---------------|------------|---------------------|------------|-------------|-----------|----------|--------|----------|-----------|--------------|------------|------|
| Government | Germany     | Carretera | None          | 1513       | 3                   | 350        | 529550      | 0         | 529550   | 393380 | 136170   | 01-Dec-14 | 12           | December   | 2014 |
| Government | Germany     | Paseo     | None          | 1006       | 10                  | 350        | 352100      | 0         | 352100   | 261560 | 90540    | 01-Jun-14 | 6            | June       | 2014 |
| Government | Canada      | Paseo     | None          | 1725       | 10                  | 350        | 603750      | 0         | 603750   | 448500 | 155250   | 01-Nov-13 | 11           | November   | 2013 |
| Government | Germany     | Paseo     | None          | 1513       | 10                  | 350        | 529550      | 0         | 529550   | 393380 | 136170   | 01-Dec-14 | 12           | December   | 2014 |
| Government | Germany     | Velo      | None          | 1006       | 120                 | 350        | 352100      | 0         | 352100   | 261560 | 90540    | 01-Jun-14 | 6            | June       | 2014 |
| Government | France      | VTT       | None          | 1527       | 250                 | 350        | 534450      | 0         | 534450   | 397020 | 137430   | 01-Sep-13 | 9            | September  | 2013 |
| Government | France      | Amarilla  | None          | 2750       | 260                 | 350        | 962500      | 0         | 962500   | 715000 | 247500   | 01-Feb-14 | 2            | February   | 2014 |
| Government | Mexico      | Carretera | Low           | 1210       | 3                   | 350        | 423500      | 4235      | 419265   | 314600 | 104665   | 01-Mar-14 | 3            | March      | 2014 |
| Government | Mexico      | Carretera | Low           | 1397       | 3                   | 350        | 488950      | 4889.5    | 484060.5 | 363220 | 120840.5 | 01-Oct-14 | 10           | October    | 2014 |
| Government | France      | Carretera | Low           | 2155       | 3                   | 350        | 754250      | 7542.5    | 746707.5 | 560300 | 186407.5 | 01-Dec-14 | 12           | December   | 2014 |
| Government | France      | Paseo     | Low           | 2155       | 10                  | 350        | 754250      | 7542.5    | 746707.5 | 560300 | 186407.5 | 01-Dec-14 | 12           | December   | 2014 |
| Government | Canada      | VTT       | Low           | 943.5      | 250                 | 350        | 330225      | 3302.25   | 326922.8 | 245310 | 81612.75 | 01-Apr-14 | 4            | April      | 2014 |
| Government | Mexico      | VTT       | Low           | 1397       | 250                 | 350        | 488950      | 4889.5    | 484060.5 | 363220 | 120840.5 | 01-Oct-14 | 10           | October    | 2014 |
| Government | Canada      | Carretera | Low           | 2852       | 3                   | 350        | 998200      | 19964     | 978236   | 741520 | 236716   | 01-Dec-14 | 12           | December   | 2014 |
| Government | Canada      | Paseo     | Low           | 2852       | 10                  | 350        | 998200      | 19964     | 978236   | 741520 | 236716   | 01-Dec-14 | 12           | December   | 2014 |
| Government | Germany     | Velo      | Low           | 2966       | 120                 | 350        | 1038100     | 20762     | 1017338  | 771160 | 246178   | 01-Oct-13 | 10           | October    | 2013 |
| Government | Germany     | Velo      | Low           | 2877       | 120                 | 350        | 1006950     | 20139     | 986811   | 748020 | 238791   | 01-Oct-14 | 10           | October    | 2014 |
| Government | Germany     | VTT       | Low           | 2877       | 250                 | 350        | 1006950     | 20139     | 986811   | 748020 | 238791   | 01-Oct-14 | 10           | October    | 2014 |
| Government | United Stat | VTT       | Low           | 266        | 250                 | 350        | 93100       | 1862      | 91238    | 69160  | 22078    | 01-Dec-13 | 12           | December   | 2013 |
| Government | Mexico      | VTT       | Low           | 1940       | 250                 | 350        | 679000      | 13580     | 665420   | 504400 | 161020   | 01-Dec-13 | 12           | December   | 2013 |
| Government | Germany     | Amarilla  | Low           | 2966       | 260                 | 350        | 1038100     | 20762     | 1017338  | 771160 | 246178   | 01-Oct-13 | 10           | October    | 2013 |
| Government | Germany     | Montana   | Low           | 1797       | 5                   | 350        | 628950      | 18868.5   | 610081.5 | 467220 | 142861.5 | 01-Sep-13 | 9            | September  | 2013 |
| Government | Mexico      | VTT       | Low           | 1642       | 250                 | 350        | 574700      | 17241     | 557459   | 426920 | 130539   | 01-Aug-14 | 8            | August     | 2014 |
| Government | United Stat | Carretera | Low           | 274        | 3                   | 350        | 95900       | 3836      | 92064    | 71240  | 20824    | 01-Dec-14 | 12           | December   | 2014 |
| Government | United Stat | Paseo     | Low           | 3450       | 10                  | 350        | 1207500     | 48300     | 1159200  | 897000 | 262200   | 01-Jul-14 | 7            | July       | 2014 |
| Government | United Stat | Paseo     | Low           | 274        | 10                  | 350        | 95900       | 3836      | 92064    | 71240  | 20824    | 01-Dec-14 | 12           | December   | 2014 |
| Government | France      | Velo      | Low           | 2177       | 120                 | 350        | 761950      | 30478     | 731472   | 566020 | 165452   | 01-Oct-14 | 10           | October    | 2014 |
| Government | United Stat | VTT       | Low           | 349        | 250                 | 350        | 122150      | 4886      | 117264   | 90740  | 26524    | 01-Sep-13 | 9            | September  | 2013 |
| Government | France      | VTT       | Low           | 2177       | 250                 | 350        | 761950      | 30478     | 731472   | 566020 | 165452   | 01-Oct-14 | 10           | October    | 2014 |
| Government | Mexico      | Amarilla  | Low           | 1865       | 260                 | 350        | 652750      | 26110     | 626640   | 484900 | 141740   | 01-Feb-14 | 2            | February   | 2014 |
| Government | Germany     | Amarilla  | Low           | 1907       | 260                 | 350        | 667450      | 26698     | 640752   | 495820 | 144932   | 01-Sep-14 | 9            | September  | 2014 |
| Government | Canada      | Amarilla  | Low           | 1778       | 260                 | 350        | 622300      | 24892     | 597408   | 462280 | 135128   | 01-Dec-13 | 12           | December   | 2013 |
| Government | France      | Montana   | Medium        | 1384.5     | 5                   | 350        | 484575      | 24228.75  | 460346.3 | 359970 | 100376.3 | 01-Jan-14 | 1            | January    | 2014 |
| Government | Mexico      | Montana   | Medium        | 720        | 5                   | 350        | 252000      | 12600     | 239400   | 187200 | 52200    | 01-Sep-13 | 9            | September  | 2013 |
| Government | United Stat | Paseo     | Medium        | 602        | 10                  | 350        | 210700      | 10535     | 200165   | 156520 | 43645    | 01-Jun-14 | 6            | June       | 2014 |

## Methodology

Step 1: Data Collection

Step 2: Data Cleaning (Python)

Step 3: SQL Analysis

Step 4: Dashboard Development (Power BI)

Step 5: Business Insights

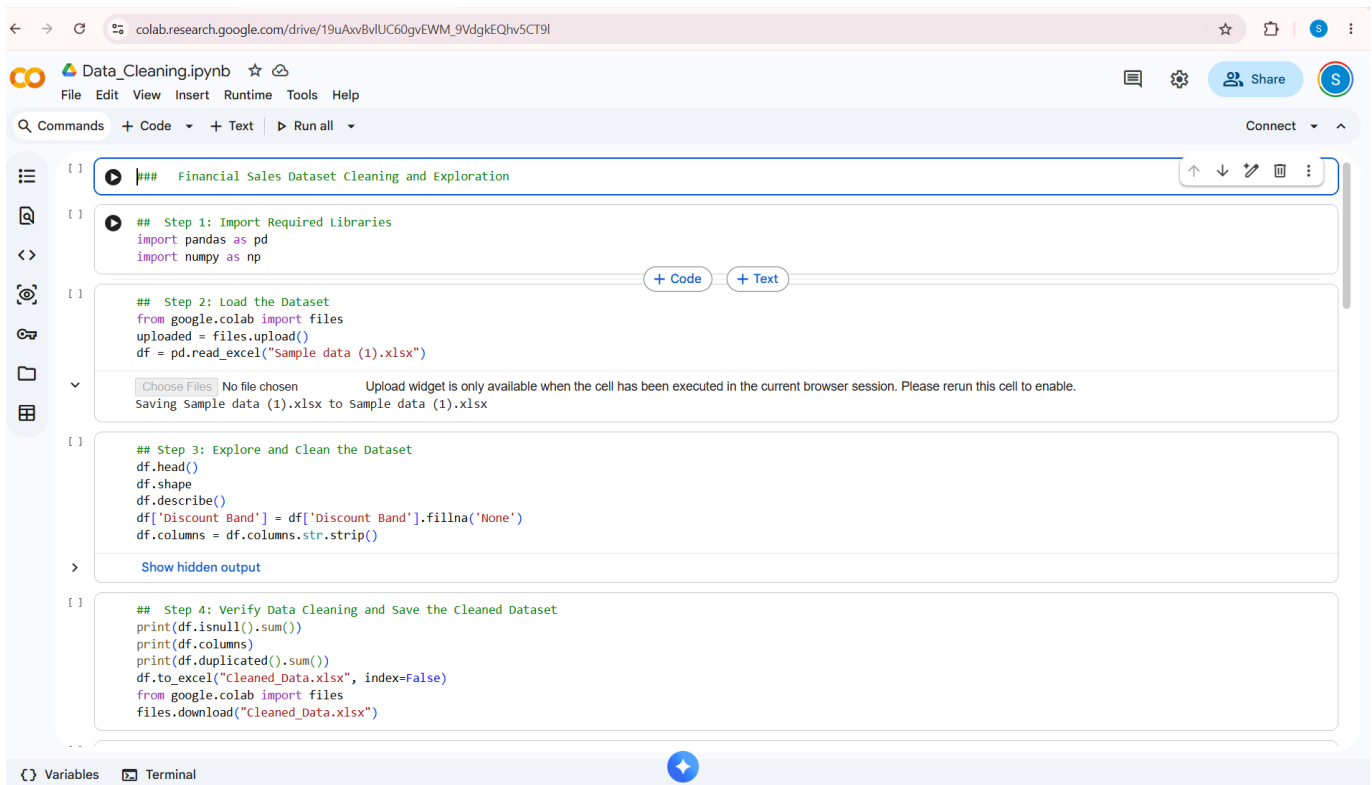
Step 6: Documentation

# CHAPTER 1

## Python Data Cleaning

### 1.1 Activities:

- Imported dataset
- Checked shape and statistics
- Identified missing values
- Replaced missing values in Discount Band with 'None'
- Verified duplicate records
- Exported cleaned dataset.



```
## Financial Sales Dataset Cleaning and Exploration

## Step 1: Import Required Libraries
import pandas as pd
import numpy as np

## Step 2: Load the Dataset
from google.colab import files
uploaded = files.upload()
df = pd.read_excel("Sample data (1).xlsx")

## Step 3: Explore and Clean the Dataset
df.head()
df.shape
df.describe()
df['Discount Band'] = df['Discount Band'].fillna('None')
df.columns = df.columns.str.strip()

## Step 4: Verify Data Cleaning and Save the Cleaned Dataset
print(df.isnull().sum())
print(df.columns)
print(df.duplicated().sum())
df.to_excel("Cleaned_Data.xlsx", index=False)
from google.colab import files
files.download("Cleaned_Data.xlsx")
```

Figure 1.1: Data Cleaning and Preprocessing Using Python

## 1.2 Overview

Python was used to clean, preprocess, and analyze the Financial Sales dataset before performing SQL analysis and creating the Power BI dashboard. The implementation was carried out in **Google Colab** using the **Pandas** and **NumPy** libraries. The objective was to ensure that the dataset was accurate, consistent, and free from missing or duplicate values, making it suitable for further analysis.

## 1.3 Tools and Libraries Used

- Python
- Google Colab
- Pandas
- NumPy

## Data Loading

The dataset was imported into Google Colab from an Excel file using the Pandas library. Initial inspection was performed to understand the dataset structure, column names, data types, and the number of records available.

## Data Exploration

Before cleaning the dataset, exploratory analysis was conducted to understand the overall quality of the data. Various functions such as `head()`, `info()`, `describe()`, and `shape` were used to examine the dataset's structure, data types, summary statistics, and total number of rows and columns.

## Handling Missing Values

The dataset contained missing values in the **Discount Band** column. These missing values were replaced with the value **"None"** using the `fillna()` function to maintain data consistency and avoid issues during analysis.

## Column Name Standardization

Extra spaces present in column names were removed using the `str.strip()` method. Standardizing column names improves readability and simplifies referencing columns during SQL queries and Power BI visualizations.

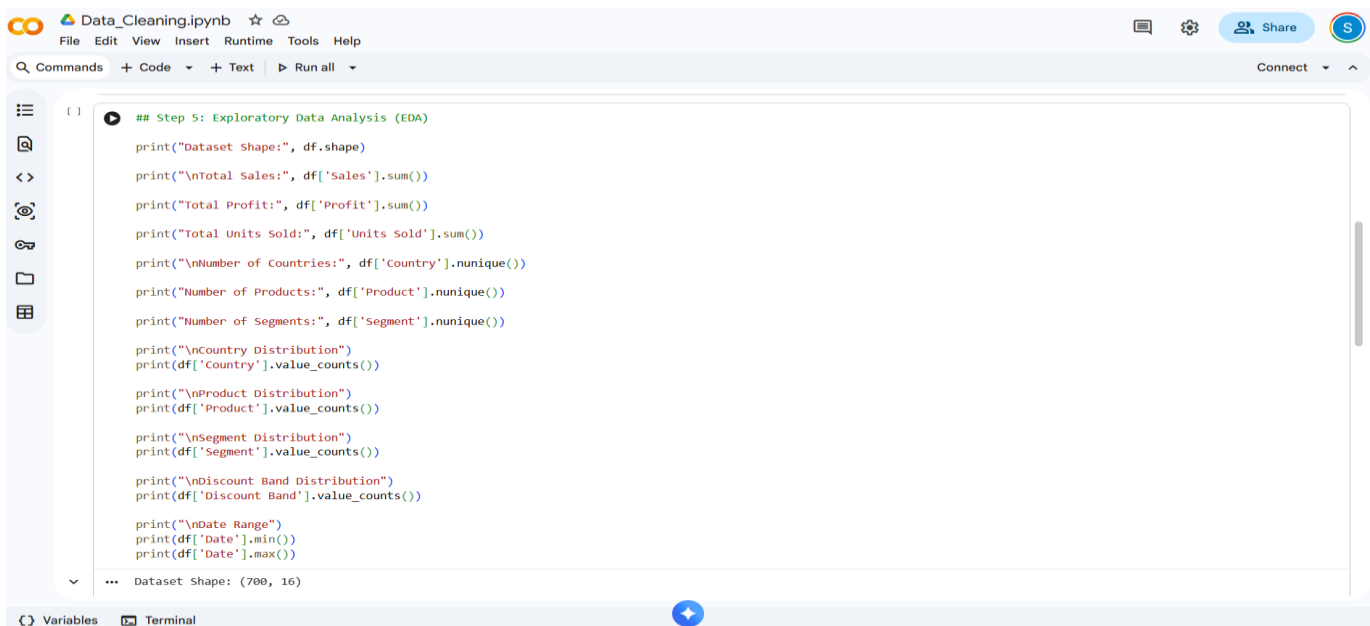
## Duplicate Record Check

The dataset was examined for duplicate records to ensure data integrity. No duplicate records were found, confirming that each record represented unique sales information.

## Data Validation

After cleaning, the dataset was validated to ensure that:

- Missing values were successfully handled.
- Duplicate records were absent.
- Column names were standardized.
- Data types remained consistent.
- The dataset was ready for further analysis.



```
## Step 5: Exploratory Data Analysis (EDA)

print("Dataset Shape:", df.shape)

print("\nTotal Sales:", df['Sales'].sum())

print("Total Profit:", df['Profit'].sum())

print("Total Units Sold:", df['Units Sold'].sum())

print("\nNumber of Countries:", df['Country'].nunique())

print("Number of Products:", df['Product'].nunique())

print("Number of Segments:", df['Segment'].nunique())

print("\nCountry Distribution")
print(df['Country'].value_counts())

print("\nProduct Distribution")
print(df['Product'].value_counts())

print("\nSegment Distribution")
print(df['Segment'].value_counts())

print("\nDiscount Band Distribution")
print(df['Discount Band'].value_counts())

print("\nDate Range")
print(df['Date'].min())
print(df['Date'].max())
```

Dataset Shape: (700, 16)

Figure 1.2: Exploratory Data Analysis Code Snippet

## Financial Sales Analysis | Internship Project

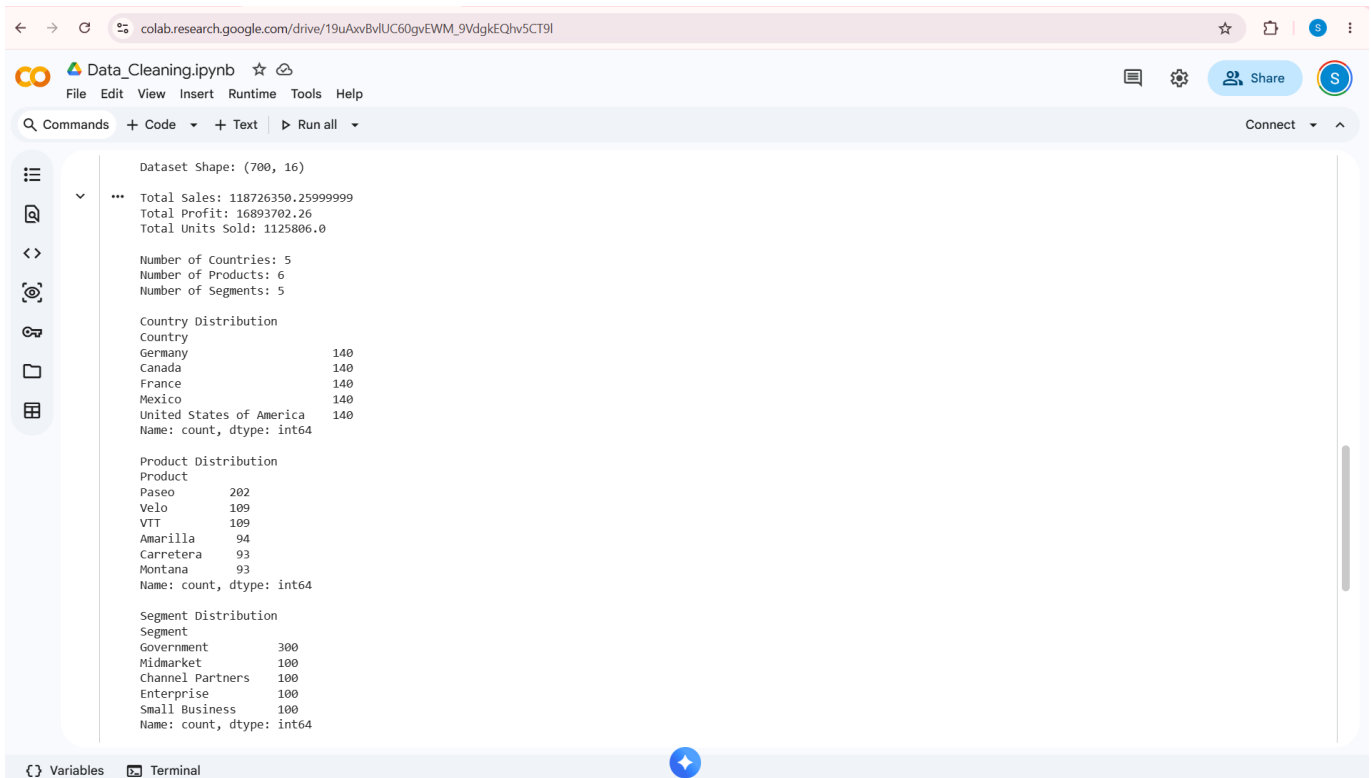


Figure 1.3: Exploratory Data Analysis Results

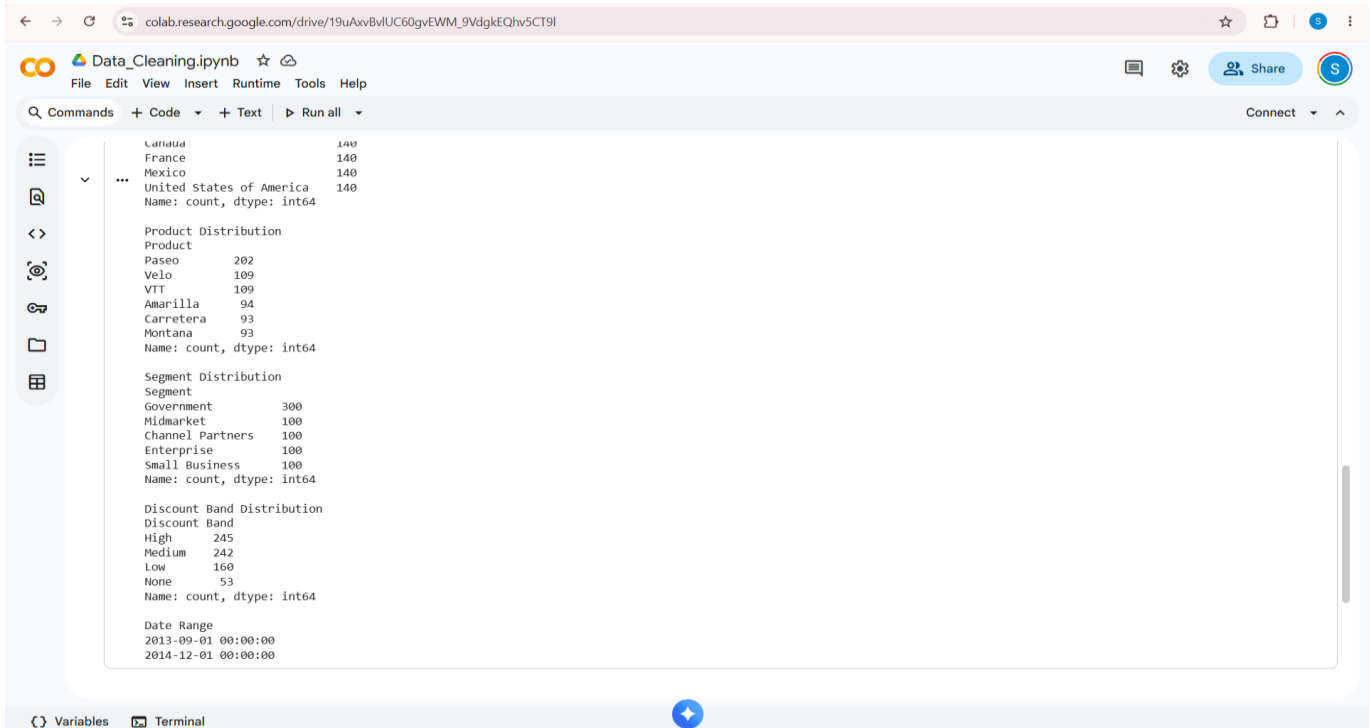


Figure 1.4 : Exploratory Data Analysis Results(continued)

## 1.4 Exploratory Data Analysis

Basic exploratory data analysis was performed to gain insights into the dataset. Key business metrics such as **Total Sales**, **Total Profit**, and **Total Units Sold** were calculated. Additionally, distributions of products, countries, customer segments, and discount bands were analyzed to better understand sales performance.

## 1.5 Outcome

The Python data cleaning process successfully transformed the raw dataset into a clean and structured format suitable for business analysis. This preprocessing stage improved data quality, minimized inconsistencies, and provided a reliable foundation for SQL querying and interactive dashboard creation.

## CHAPTER 2

### SQL Data Analysis

#### 2.1 Overview

After completing the data cleaning process in Python, the cleaned dataset was imported into **MySQL Workbench** for performing Structured Query Language (SQL) analysis. SQL was used to retrieve meaningful business insights from the dataset by executing various analytical queries. These queries helped summarize sales performance, identify top-performing products and regions, and analyze revenue trends across different business segments.

#### 2.2 Database Setup

The cleaned dataset (**Cleaned\_Data.xlsx**) was converted into CSV format and imported into MySQL Workbench. A database named **financial\_sales** was created, and the dataset was stored in a table named **financial\_data**. This enabled efficient querying and analysis using SQL commands.

#### 2.3 SQL Queries Performed

A series of SQL queries were executed to analyze the financial sales data. The analysis included:

- Viewing the complete dataset.
- Counting the total number of records.
- Calculating total sales.
- Calculating total profit.
- Calculating total units sold.
- Analyzing sales across different countries.
- Identifying profit generated by each product.
- Comparing sales across customer segments.

- Analyzing monthly sales trends.
- Identifying the top five products based on profit.

Each query was designed to answer a specific business question and provide valuable insights into overall sales performance.

## SQL Query Screenshots

The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'SCHEMAS' tree with 'financial\_sales' selected. The main window shows a SQL query in the 'Query 1' tab:

```

1 • CREATE DATABASE financial_sales;
2 • USE financial_sales;
3
4 #Query 1: View the Dataset:
5 • SELECT * FROM financial_data;
6

```

The 'Result Grid' shows the output of the query, displaying 20 rows of data. The columns are: Segment, Country, Product, Discount Band, Units Sold, Manufacturing Price, Sale Price, Gross Sales, Discounts, Sales, COGS, Profit, Date, Month Number, Month Name, and Year.

| Segment    | Country     | Product   | Discount Band | Units Sold | Manufacturing Price | Sale Price | Gross Sales | Discounts | Sales   | COGS   | Profit | Date                | Month Number | Month Name | Year |
|------------|-------------|-----------|---------------|------------|---------------------|------------|-------------|-----------|---------|--------|--------|---------------------|--------------|------------|------|
| Government | Germany     | Carretera | None          | 1513       | 3                   | 350        | 529550      | 0         | 529550  | 393380 | 136170 | 2014-12-01 00:00:00 | 12           | December   | 2014 |
| Government | Germany     | Paseo     | None          | 1006       | 10                  | 350        | 352100      | 0         | 352100  | 261560 | 90540  | 2014-06-01 00:00:00 | 6            | June       | 2014 |
| Government | Canada      | Paseo     | None          | 1725       | 10                  | 350        | 603750      | 0         | 603750  | 448500 | 155250 | 2013-11-01 00:00:00 | 11           | November   | 2013 |
| Government | Germany     | Paseo     | None          | 1513       | 10                  | 350        | 529550      | 0         | 529550  | 393380 | 136170 | 2014-12-01 00:00:00 | 12           | December   | 2014 |
| Government | Germany     | Velo      | None          | 1006       | 120                 | 350        | 352100      | 0         | 352100  | 261560 | 90540  | 2014-06-01 00:00:00 | 6            | June       | 2014 |
| Government | France      | VTT       | None          | 1527       | 250                 | 350        | 534450      | 0         | 534450  | 397020 | 137430 | 2013-09-01 00:00:00 | 9            | September  | 2013 |
| Government | France      | Amarilla  | None          | 2750       | 260                 | 350        | 962500      | 0         | 962500  | 715000 | 247500 | 2014-02-01 00:00:00 | 2            | February   | 2014 |
| Government | Mexico      | Carretera | Low           | 1210       | 3                   | 350        | 423500      | 4235      | 419265  | 314600 | 104665 | 2014-03-01 00:00:00 | 3            | March      | 2014 |
| Government | France      | Carretera | Low           | 1397       | 3                   | 350        | 488950      | 4890      | 484061  | 363220 | 120841 | 2014-10-01 00:00:00 | 10           | October    | 2014 |
| Government | France      | Carretera | Low           | 2155       | 3                   | 350        | 754250      | 7543      | 746708  | 560300 | 186408 | 2014-12-01 00:00:00 | 12           | December   | 2014 |
| Government | France      | Paseo     | Low           | 2155       | 10                  | 350        | 754250      | 7543      | 746708  | 560300 | 186408 | 2014-12-01 00:00:00 | 12           | December   | 2014 |
| Government | Canada      | VTT       | Low           | 944        | 250                 | 350        | 330225      | 3302      | 326923  | 245310 | 81613  | 2014-04-01 00:00:00 | 4            | April      | 2014 |
| Government | Mexico      | VTT       | Low           | 1397       | 250                 | 350        | 488950      | 4890      | 484061  | 363220 | 120841 | 2014-10-01 00:00:00 | 10           | October    | 2014 |
| Government | Canada      | Carretera | Low           | 2852       | 3                   | 350        | 998200      | 19964     | 978236  | 741520 | 236716 | 2014-12-01 00:00:00 | 12           | December   | 2014 |
| Government | Canada      | Paseo     | Low           | 2852       | 10                  | 350        | 998200      | 19964     | 978236  | 741520 | 236716 | 2014-12-01 00:00:00 | 12           | December   | 2014 |
| Government | Germany     | Velo      | Low           | 2966       | 120                 | 350        | 1038100     | 20762     | 1017338 | 771160 | 246178 | 2013-10-01 00:00:00 | 10           | October    | 2013 |
| Government | Germany     | Velo      | Low           | 2877       | 120                 | 350        | 1006950     | 20139     | 986811  | 748020 | 238791 | 2014-10-01 00:00:00 | 10           | October    | 2014 |
| Government | Germany     | VTT       | Low           | 2877       | 250                 | 350        | 1006950     | 20139     | 986811  | 748020 | 238791 | 2014-10-01 00:00:00 | 10           | October    | 2014 |
| Government | United S... | VTT       | Low           | 266        | 250                 | 350        | 93100       | 1862      | 91238   | 69160  | 22078  | 2013-12-01 00:00:00 | 12           | December   | 2013 |
| Government | Mexico      | VTT       | Low           | 1940       | 250                 | 350        | 679000      | 13580     | 665420  | 504400 | 161020 | 2013-12-01 00:00:00 | 12           | December   | 2013 |
| Government | Germany     | Amarilla  | Low           | 2966       | 260                 | 350        | 1038100     | 20762     | 1017338 | 771160 | 246178 | 2013-10-01 00:00:00 | 10           | October    | 2013 |
| Government | Germany     | Montana   | Low           | 1797       | 5                   | 350        | 628950      | 18869     | 610082  | 467220 | 143862 | 2013-09-01 00:00:00 | 9            | September  | 2013 |
| Government | Mexico      | VTT       | Low           | 1642       | 250                 | 350        | 574700      | 17241     | 557459  | 426920 | 130539 | 2014-08-01 00:00:00 | 8            | August     | 2014 |
| Government | United S... | Carretera | Low           | 274        | 3                   | 350        | 95900       | 3836      | 92064   | 71240  | 20824  | 2014-12-01 00:00:00 | 12           | December   | 2014 |

Figure 2.1: Financial Sales Table in MySQL Workbench

## Financial Sales Analysis | Internship Project

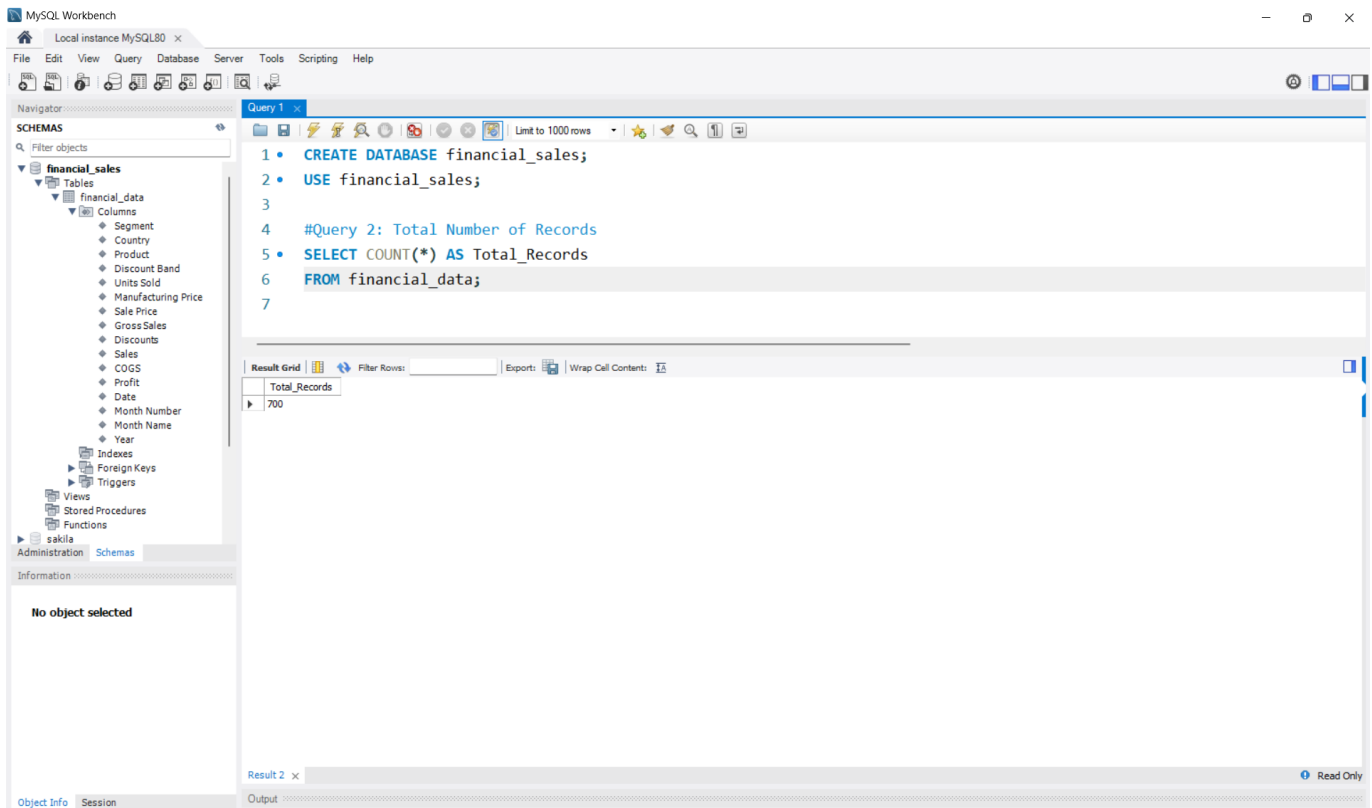


Figure 2.2 : SQL Query for Total Records

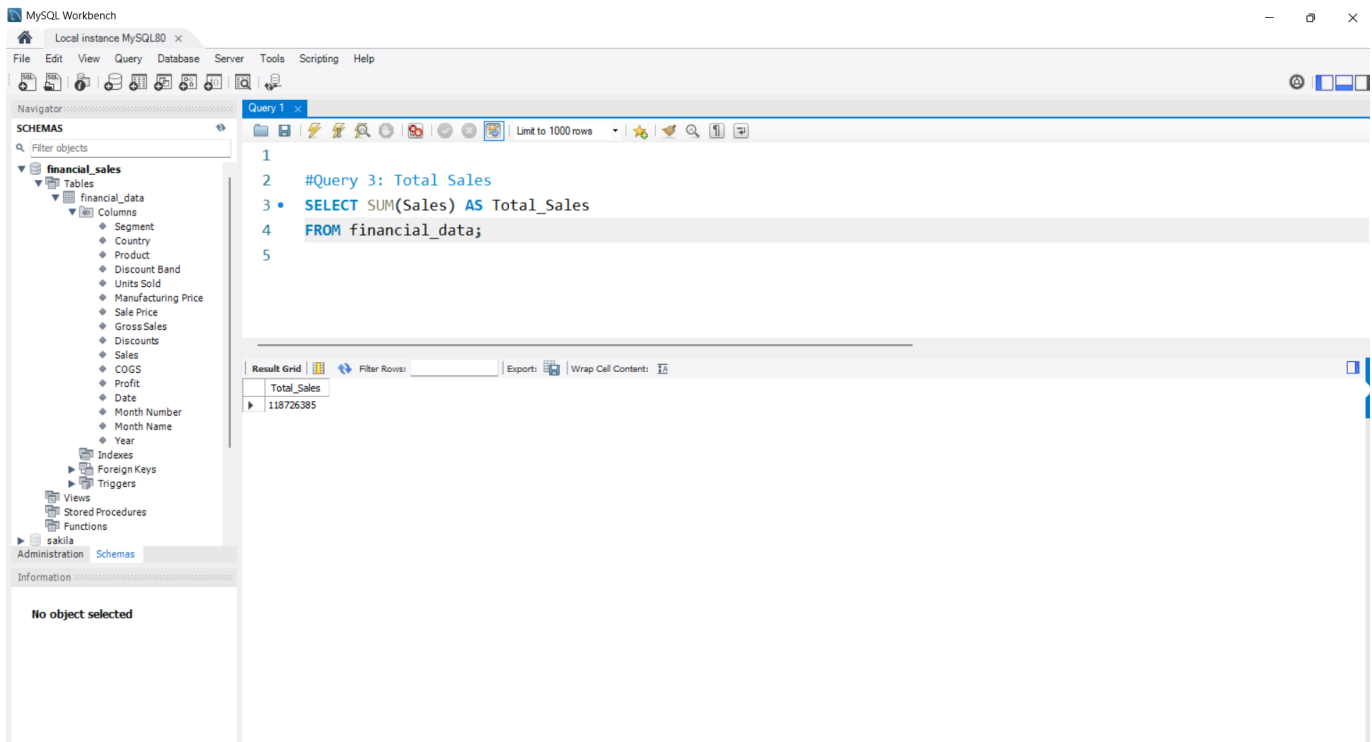


Figure 2.3 : SQL Query for Total Sales

## Financial Sales Analysis | Internship Project

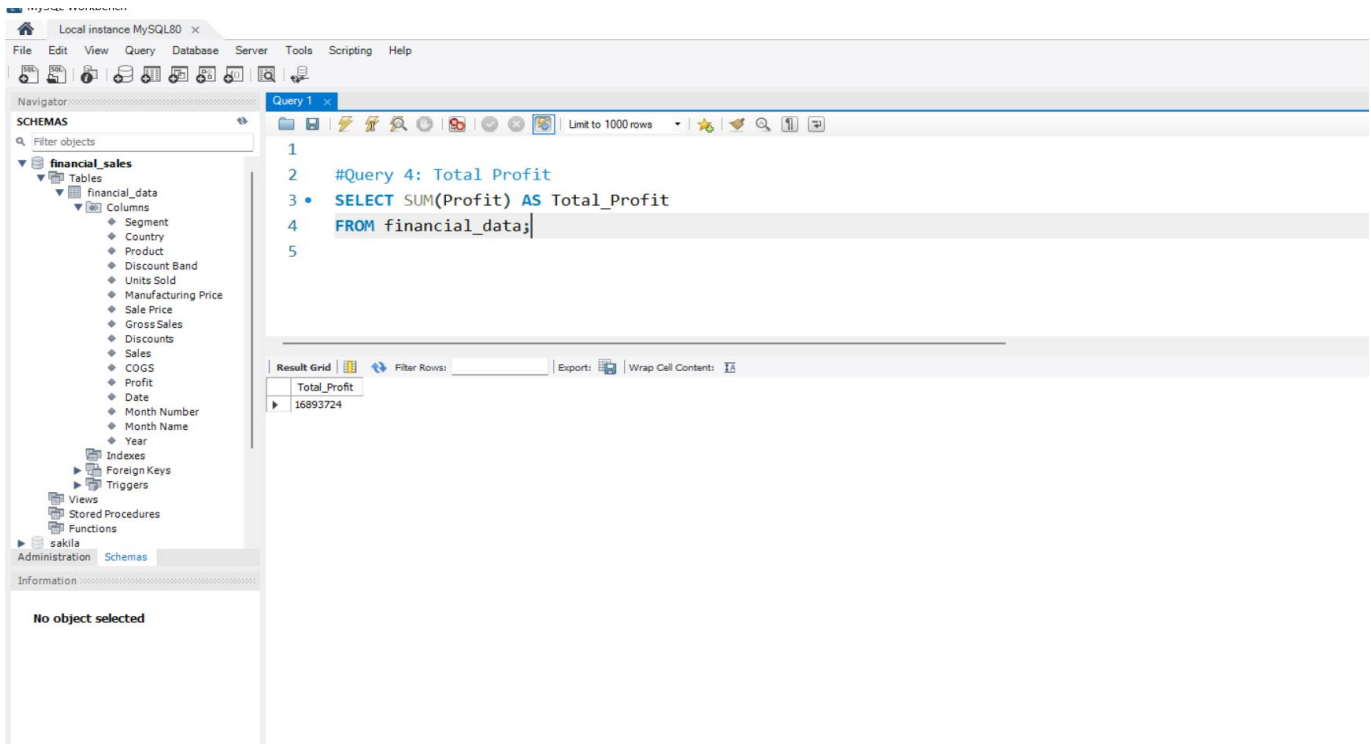


Figure 2.4 : SQL Query for Total Profit

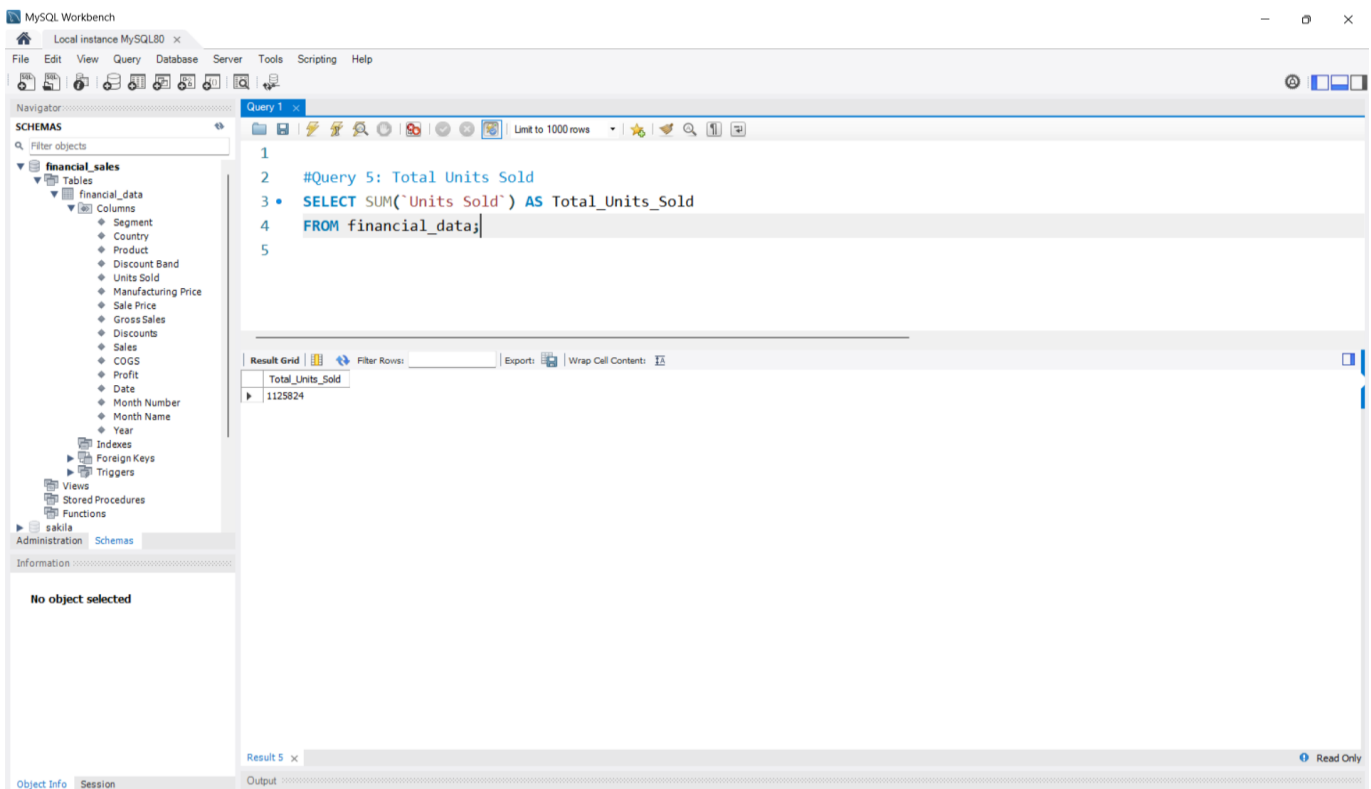


Figure 2.5 : SQL Query for Total Units Sold

## Financial Sales Analysis | Internship Project

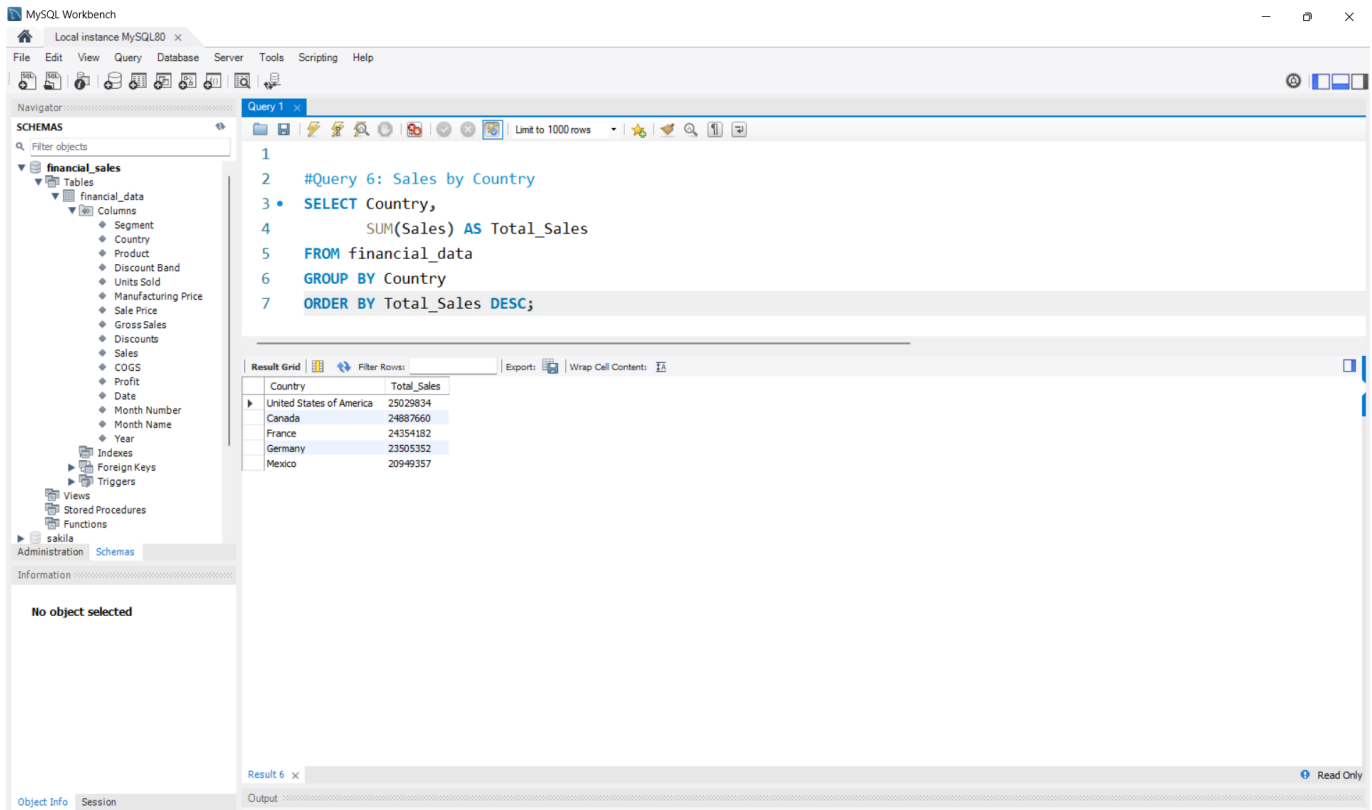


Figure 2.6 : SQL Query for Sales by Country

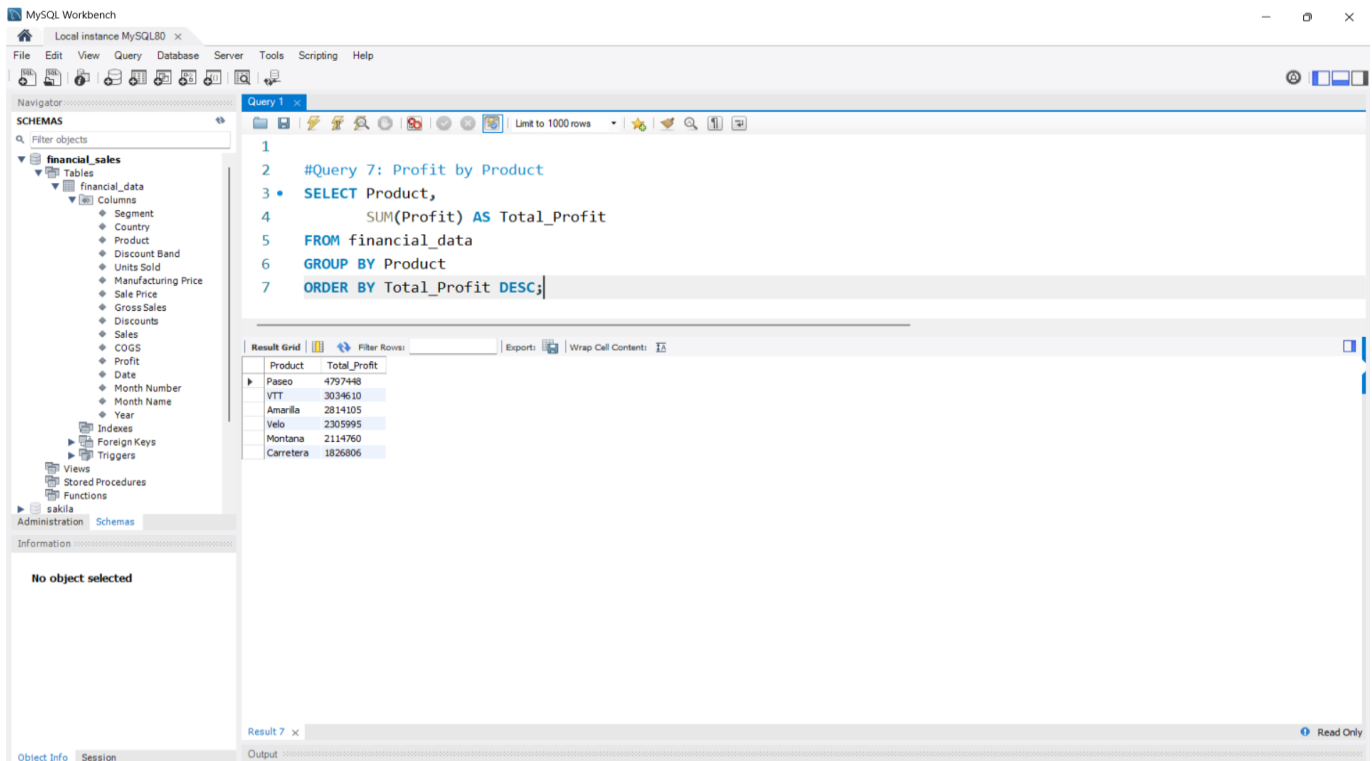


Figure 2.7 : SQL Query for Profit by Product

## Financial Sales Analysis | Internship Project

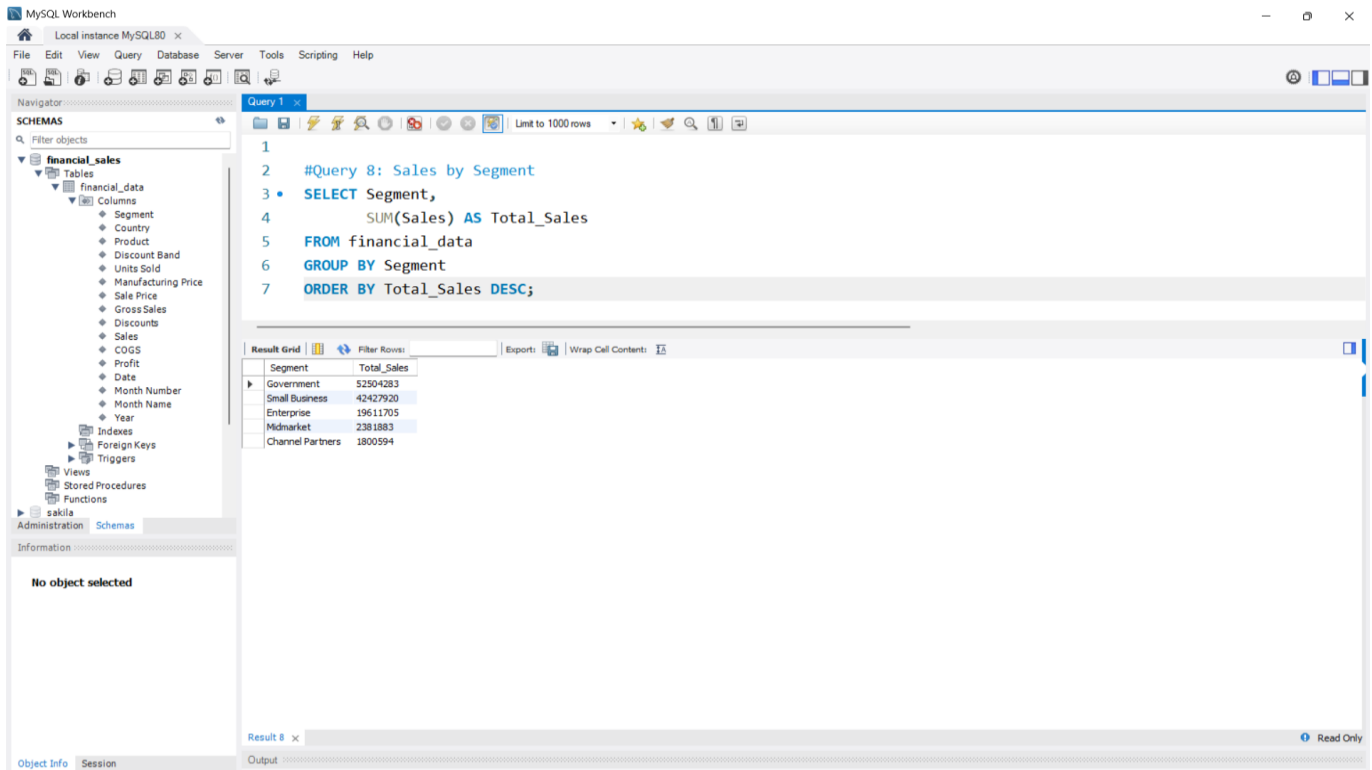


Figure 2.8 : SQL Query for Sales by Customer Segment

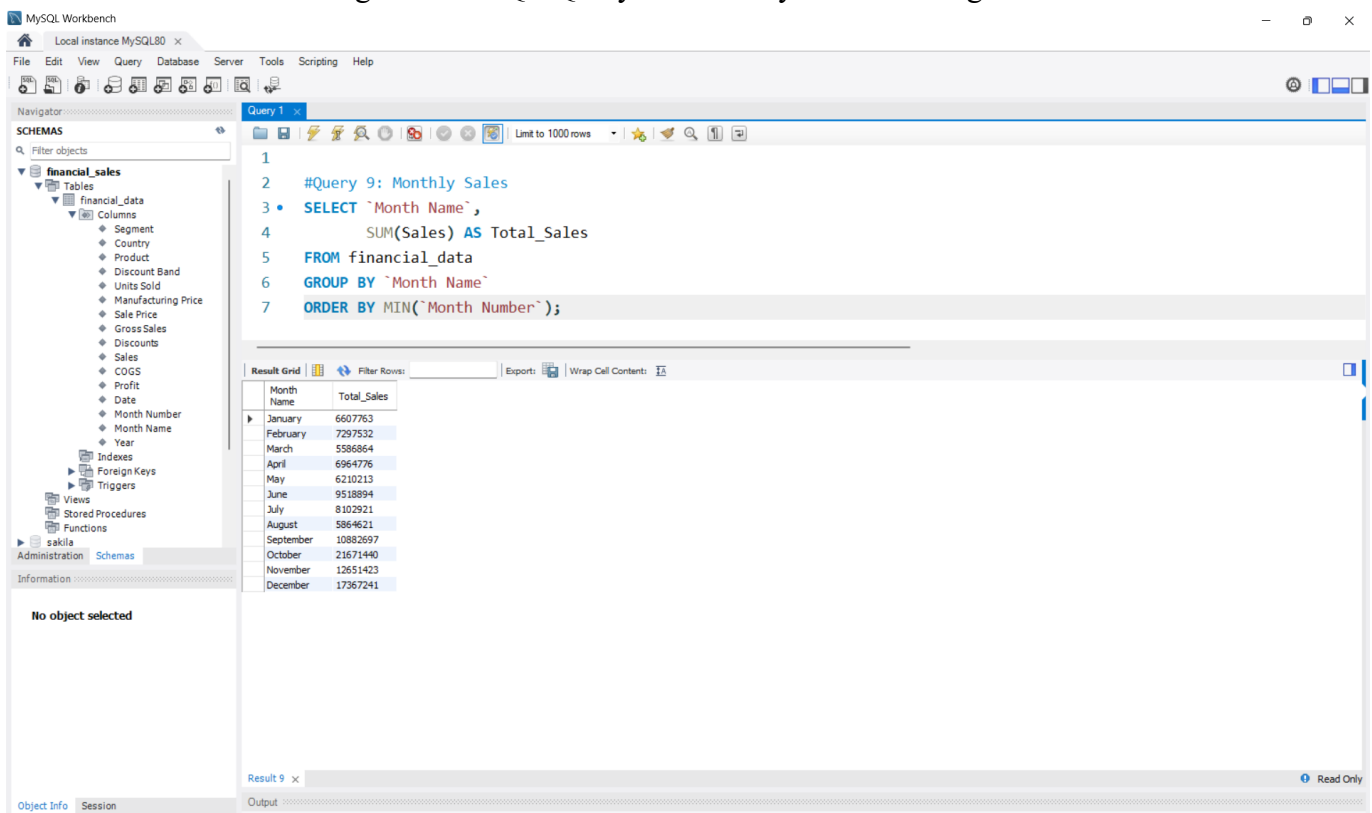


Figure 2.9 : SQL Query for Monthly Sales Analysis

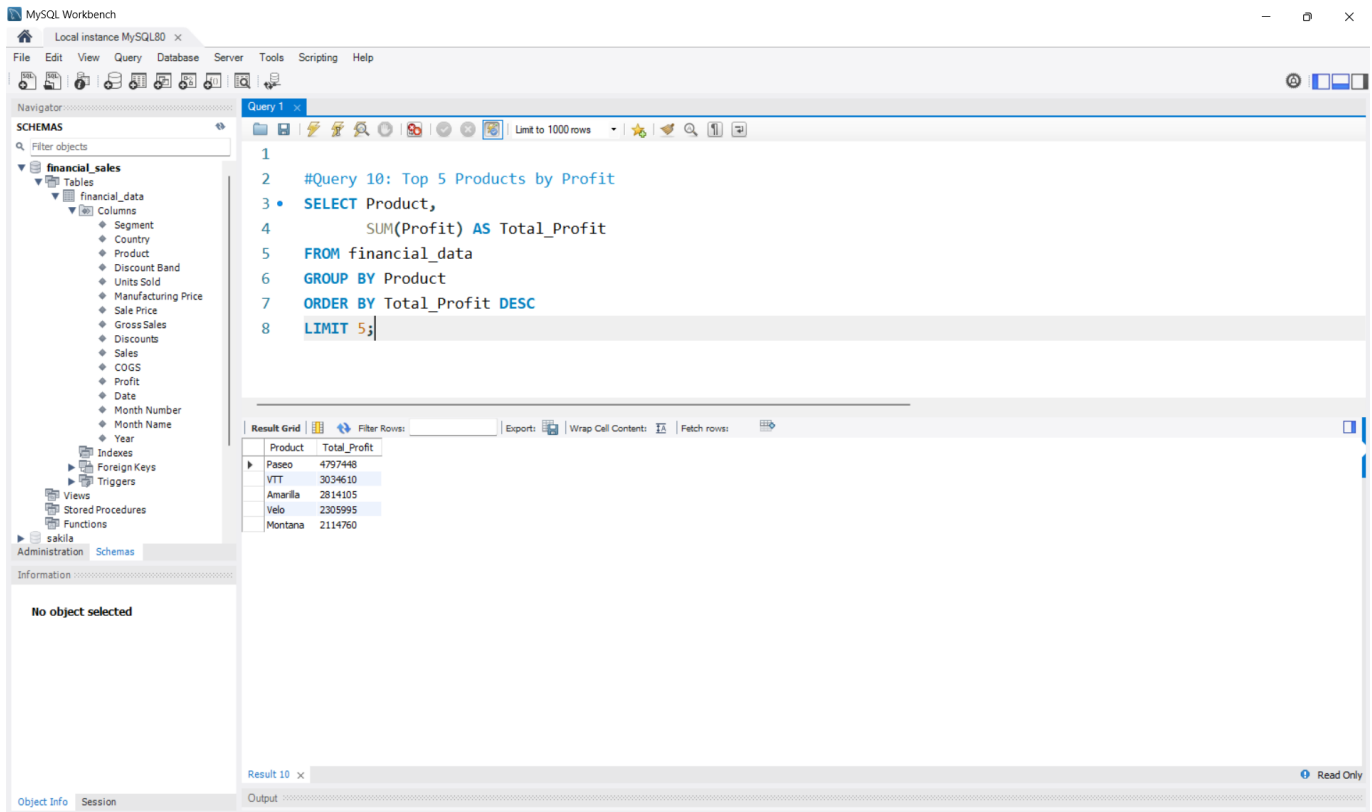


Figure 2.10 : SQL Query for Top Five Products by Profit

## 2.4 Key Findings

The SQL analysis produced several important business insights:

- The dataset contains **700** sales records.
- Total Sales amounted to **118,726,385**.
- Total Profit generated was **16,893,724**.
- Total Units Sold were **1,125,824**.
- The **United States of America** recorded the highest sales.
- The **Government** segment contributed the highest revenue.
- **Paseo** emerged as the most profitable product.

These findings provided a clear understanding of business performance and supported data-driven decision-making.

## **2.5 Business Analysis**

SQL enabled efficient extraction of meaningful information from the dataset through aggregation, grouping, sorting, and filtering operations. The analysis helped identify high-performing products, profitable customer segments, regional sales performance, and monthly sales patterns. These insights can assist organizations in optimizing sales strategies, improving profitability, and making informed business decisions.

## **2.6 Outcome**

The SQL analysis successfully transformed the cleaned dataset into actionable business insights. The results obtained from SQL served as the foundation for developing an interactive Power BI dashboard, allowing stakeholders to visualize key performance indicators and monitor business performance more effectively.

## CHAPTER 3

### Power BI Dashboard

#### 3.1 Overview

After completing data cleaning in Python and business analysis using SQL, the cleaned dataset was imported into **Microsoft Power BI** to create an interactive dashboard. Power BI was used to transform the analyzed data into meaningful visualizations, enabling better understanding of business performance and supporting data-driven decision-making. The dashboard provides an intuitive representation of key business metrics through charts, graphs, KPI cards, and slicers.

#### 3.2 Objectives of the Dashboard

The main objectives of developing the Power BI dashboard are:

- To visualize key financial performance indicators.
- To monitor total sales, profit, and units sold.
- To compare sales across different countries and customer segments.
- To identify the most profitable products.
- To analyze monthly sales trends.
- To provide interactive filtering using slicers for better business analysis.

#### 3.3 Dashboard Development

The cleaned dataset (**Cleaned\_Data.xlsx**) was imported into Microsoft Power BI. Necessary data validation was performed before creating visualizations. DAX measures were created to calculate important business metrics such as **Total Sales**, **Total Profit**, and **Total Units Sold**. Various charts and KPI cards were then designed to present the data in an interactive and user-friendly manner.

#### 3.4 Dashboard Components

The dashboard includes the following visualizations:

- KPI Cards displaying Total Sales, Total Profit, and Total Units Sold.
- Bar Chart showing Sales by Country.
- Bar Chart showing Profit by Product.
- Line Chart representing Monthly Sales Trend.
- Funnel Chart displaying Sales by Customer Segment.
- Column Chart illustrating Sales by Discount Band.
- Country Slicer for filtering data by region.
- Product Slicer for filtering data by product.

Each visualization was designed to provide meaningful insights into different aspects of the financial sales dataset.

### **3.5 Dashboard Screenshots**

The following screenshots present the completed Power BI dashboard and its visualizations. Each visual highlights a specific business metric and enables users to analyze sales performance efficiently through interactive filtering and graphical representation.

## Financial Sales Analysis | Internship Project

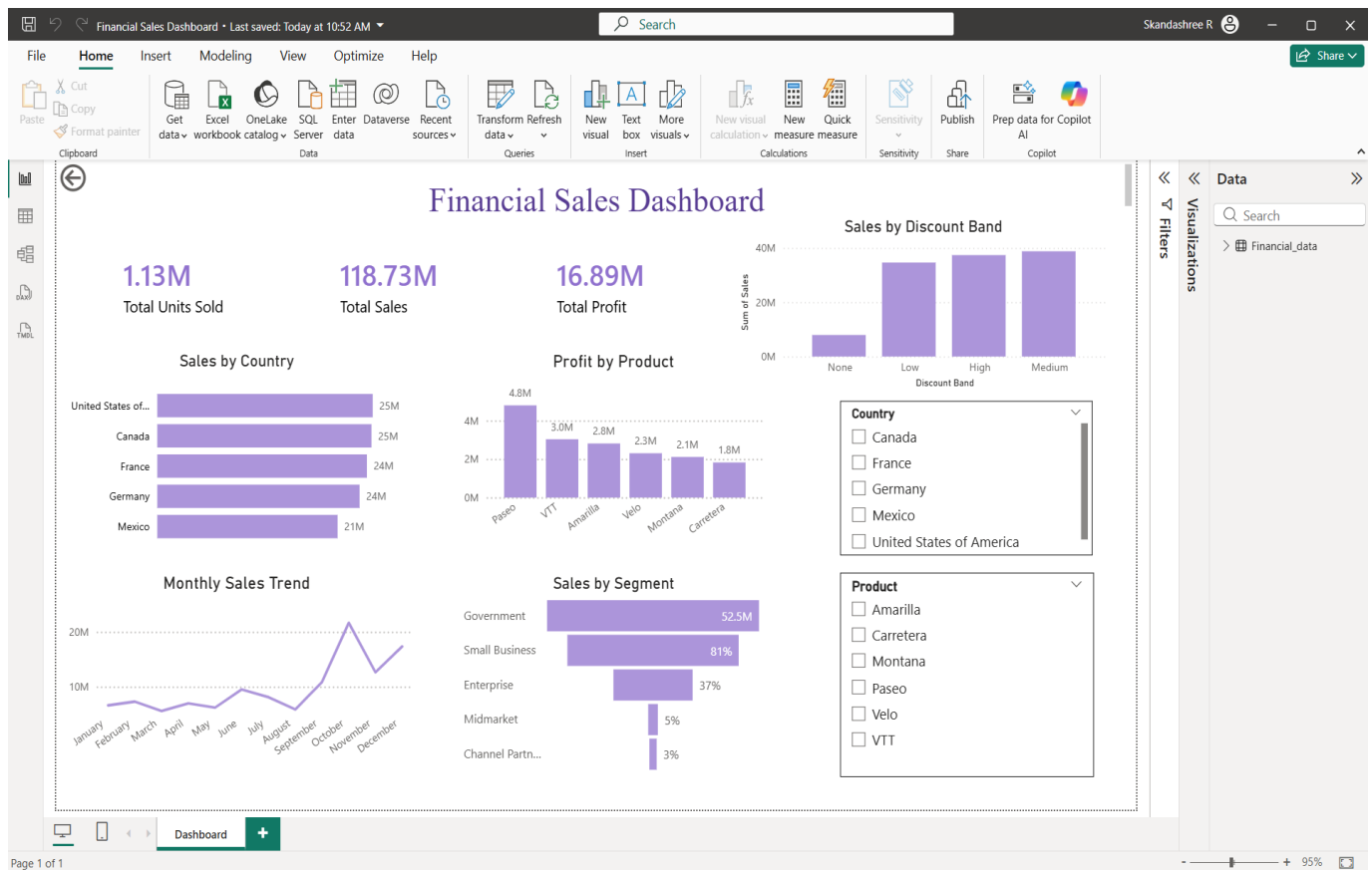


Figure 3.1: Financial Sales Dashboard Overview

**After applying Slicer:**

## Financial Sales Analysis | Internship Project

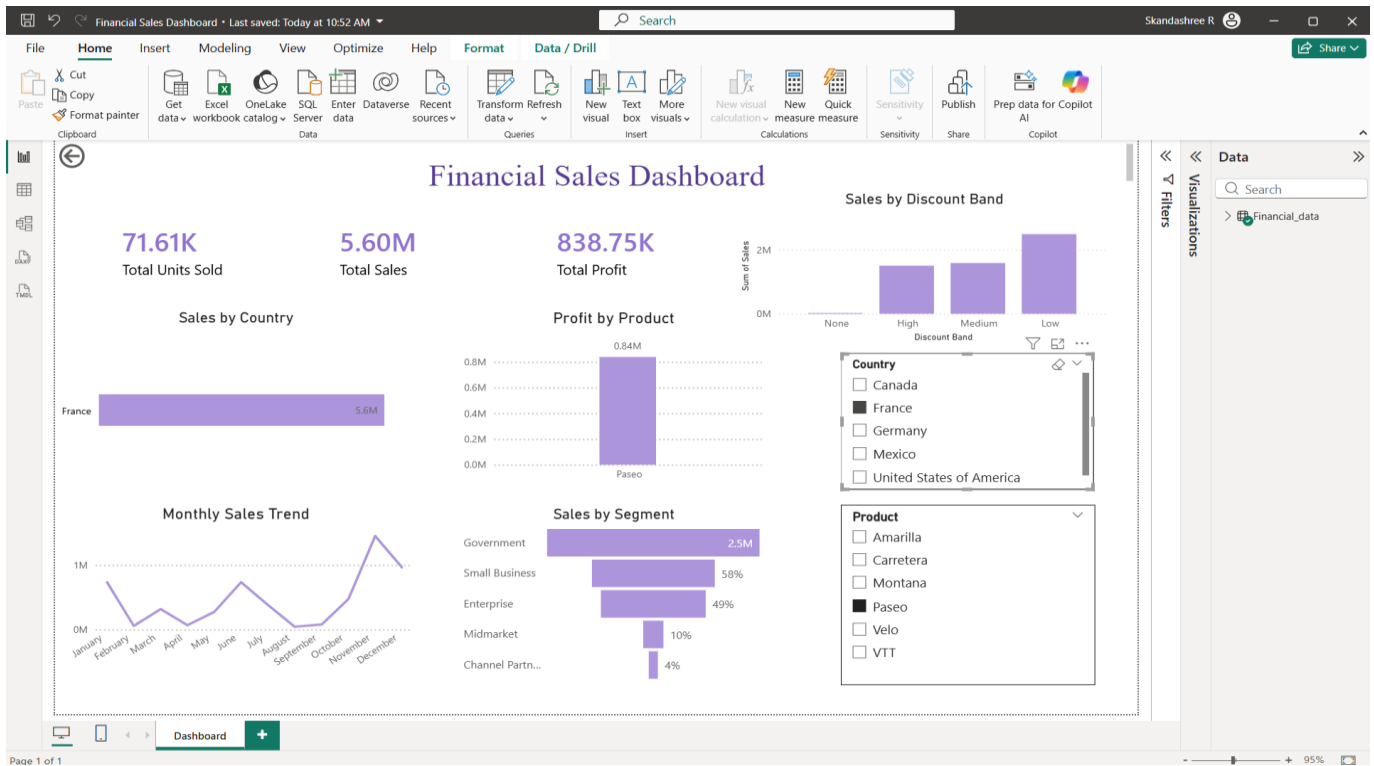


Figure 3.2: Dashboard Analysis Using Country Slicer

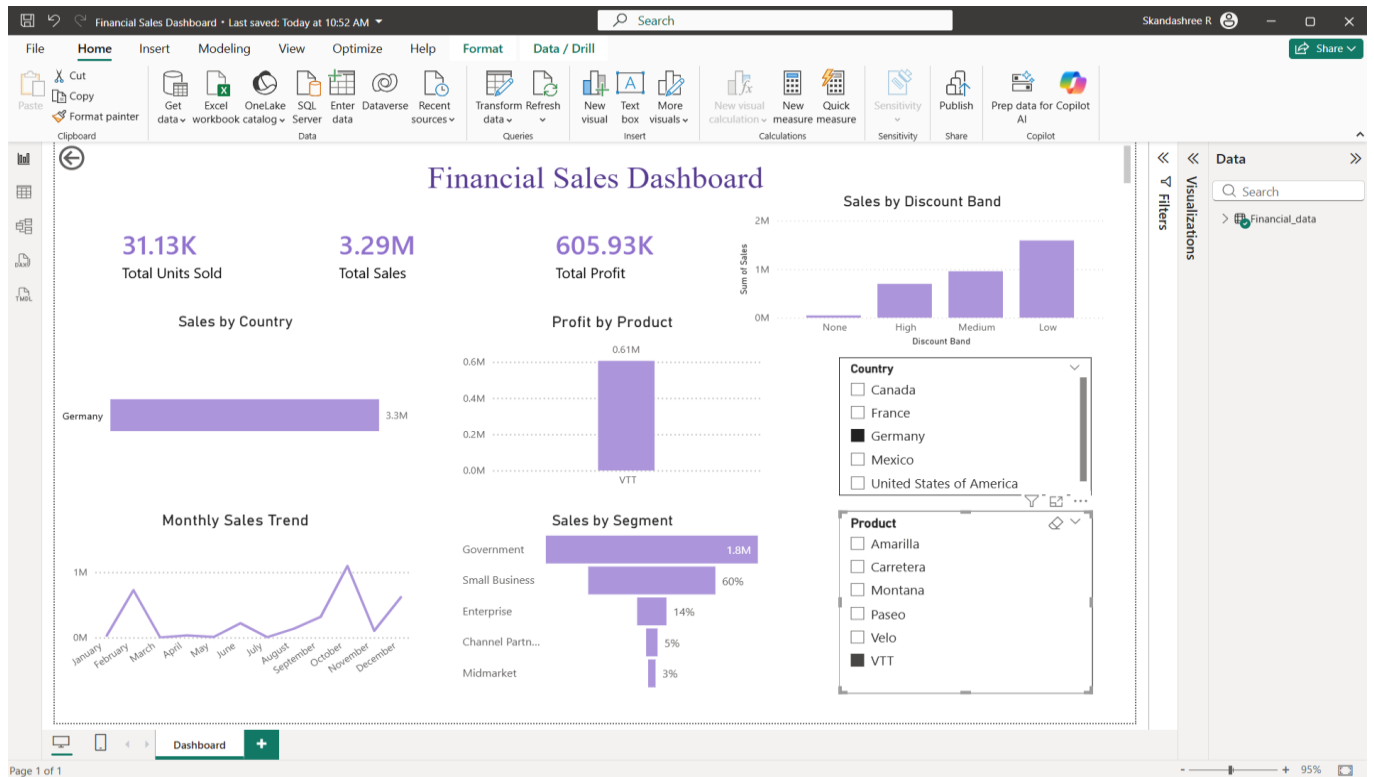


Figure 3.3: Dashboard Analysis Using Product Slicer

### 3.6 Key Insights

The Power BI dashboard provides several valuable business insights:

- The **Government** segment contributed the highest sales.
- The **United States of America** generated the highest revenue among all countries.
- **Paseo** emerged as the most profitable product.
- Sales showed noticeable variation across different months, indicating seasonal business trends.
- Interactive slicers allow users to filter and analyze data dynamically based on country and product.
- KPI cards provide a quick summary of overall business performance.

### 3.7 Outcome

The Power BI dashboard successfully transformed raw analytical results into interactive visualizations that are easy to interpret and analyze. It enables stakeholders to monitor key performance indicators, identify sales trends, evaluate product performance, and make informed business decisions. The dashboard demonstrates the effectiveness of Power BI as a business intelligence tool for presenting complex datasets in a clear and visually appealing manner.

## CHAPTER 4

### Business Insights

#### 4.1 Overview

The analysis performed using Python, SQL, and Power BI provided valuable insights into the Financial Sales dataset. Data cleaning ensured the accuracy and consistency of the dataset, SQL queries extracted meaningful business information, and Power BI transformed the results into interactive visualizations. These findings help organizations understand their sales performance, identify profitable business areas, and make informed strategic decisions.

#### 4.2 Key Business Insights

##### 1. Overall Sales Performance

The organization generated a total sales revenue of **118.73 Million**, indicating a strong market presence and consistent business operations during the analyzed period.

##### 2. Profit Analysis

The total profit earned was approximately **16.89 Million**, demonstrating that the business maintained healthy profitability while managing operational costs effectively.

##### 3. Product Performance

Among all products, **Paseo** generated the highest profit, making it the best-performing product in the portfolio. This suggests strong customer demand and effective pricing strategies.

##### 4. Country-wise Sales

The **United States of America** recorded the highest sales, indicating it is the organization's most significant revenue-generating market. Other countries also contributed to overall business growth but at comparatively lower levels.

## **5. Customer Segment Analysis**

The **Government** segment contributed the highest revenue among all customer segments. This indicates that government-based customers play a crucial role in the organization's overall sales performance.

## **6. Monthly Sales Trend**

Monthly sales analysis revealed fluctuations throughout the year, with higher sales observed during specific months. This pattern can help businesses identify seasonal demand and improve future sales planning.

## **7. Discount Analysis**

The analysis of discount bands showed that offering moderate discounts contributed positively to sales performance. Appropriate discount strategies can improve customer acquisition without significantly reducing profitability.

## **8. Dashboard Effectiveness**

The interactive Power BI dashboard enables users to monitor business performance efficiently through KPI cards, charts, and slicers. Decision-makers can easily filter information by country and product, making the dashboard a valuable business intelligence tool.

## CHAPTER 5

### Recommendations

Based on the analysis performed, the following recommendations are suggested:

- Increase focus on high-performing products such as **Paseo** to maximize profitability.
- Strengthen marketing efforts in countries with high revenue while exploring opportunities in lower-performing regions.
- Continue building relationships with Government customers, as they contribute significantly to overall sales.
- Optimize discount strategies by balancing customer attraction and profit margins.
- Utilize interactive dashboards for continuous monitoring of key performance indicators.
- Implement predictive analytics to forecast future sales trends and improve business planning.

## CHAPTER 6

### Future Scope

This project can be further enhanced through the following improvements:

- Integration with real-time business databases.
- Deployment of dashboards using Power BI Service for online access.
- Development of automated data refresh mechanisms.
- Implementation of Machine Learning models for sales forecasting.
- Integration with cloud platforms for scalable data storage and analytics.
- Inclusion of customer behavior analysis and predictive business intelligence techniques.

### Conclusion

This project successfully demonstrated the complete data analytics workflow using Python, SQL, and Microsoft Power BI. Python was used for data cleaning and preprocessing, SQL was utilized to perform business analysis and extract meaningful insights, and Power BI was employed to create an interactive dashboard for data visualization. The project highlighted the importance of data analytics in transforming raw data into valuable business insights, enabling organizations to make informed decisions and improve overall business performance. The skills acquired through this project provide a strong foundation for solving real-world business problems using modern data analytics tools.

## References

1. Microsoft Power BI Documentation
2. MySQL Official Documentation
3. Python Official Documentation
4. Pandas Documentation
5. NumPy Documentation
6. Microsoft Learn – Power BI
7. Google Colab Documentation